VPD Regulator

Setup & Design Guide

Automated Climate Control for Small Controlled-Environment Grow Spaces

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1. Introduction

This document is your complete guide to purchasing, assembling, and configuring the VPD Regulator system for a small, controlled-environment grow space — commonly known as a grow box or grow tent. By the end of this guide you will have a fully automated climate controller that continuously reads temperature and humidity, calculates Vapor Pressure Deficit in real time, and switches your humidifier or dehumidifier on and off automatically to hit the optimal target for each stage of your plants' life cycle.

> 💡 **Who is this guide for?**
>
> Home growers using a small tent (60×60 cm up to 120×120 cm) or a dedicated grow cabinet.
>
> Anyone who wants professional-grade climate automation without writing a single line of code.
>
> No prior experience with Home Assistant, Raspberry Pi, or Docker is required — we walk you through every step.

1.1 What is VPD and Why Does it Matter?

Vapor Pressure Deficit (VPD) is the difference between the amount of moisture the air can hold at saturation and the amount it currently holds. It is measured in kilopascals (kPa). Plants use VPD as a signal to open or close their stomata — the microscopic pores through which they exchange gases and transpire water.

When VPD is too low (air is very humid), stomata close and the plant stops transpiring. This slows nutrient uptake and creates conditions for mold and mildew. When VPD is too high (air is very dry), plants close their stomata to conserve water, limiting photosynthesis and growth. Staying inside the optimal VPD band for each growth stage is one of the most impactful things you can do to maximise yield, quality, and plant health.

Growth Stage	VPD Target (kPa)	Humidity Range	Notes
Seedling / Clone	0.4 – 0.6	~80%	Gentle, very humid
Early Vegetative	0.6 – 0.8	70–75%	Building root structure
Late Vegetative	0.8 – 1.0	65–70%	Active canopy growth
Early Flowering	1.0 – 1.2	55–65%	Transition phase
Late Flowering	1.2 – 1.4	45–55%	Prevent mold/mildew
Harvest / Drying	1.4 – 1.6	40–50%	Low humidity target

1.2 What This System Does

The VPD Regulator automates the entire process. You set a target VPD stage (e.g. "Late Vegetative") on a dashboard, and the system:

- Reads temperature and humidity from a sensor inside your grow space every 30 minutes (configurable).
- Applies the Magnus-Tetens formula to calculate the exact relative humidity needed to hit your target VPD at the current temperature.
- Compares the measured humidity against the calculated target.
- Switches your humidifier on if humidity is more than 2 % below target.
- Switches your dehumidifier on if humidity is more than 3 % above target.
- Does nothing if humidity is already within the dead band — preventing constant short cycling.

1.3 Why Home Assistant on a Raspberry Pi?

Home Assistant is an open-source home automation platform with over 3 000 integrations, an active community, and a proven track record in controlled-environment agriculture. We chose it for three reasons:

- It runs entirely on your local network — no cloud subscription, no data leaving your home.
- It has first-class support for the Zigbee sensor protocol, which offers the best price-to-quality ratio for grow room sensors.
- The automation engine is powerful enough for dead-band humidity control, VPD maths, alerts, and logging — all in a single YAML file.

A Raspberry Pi 4 (or 5) is the reference hardware because it is inexpensive (~€60-80), fanless-capable, consumes only 3–5 W at idle, and the Home Assistant Operating System (HAOS) image targets it directly — meaning installation is a one-command flash, not a manual Linux setup.

2. Hardware Shopping List

The table below lists every physical component you need. Prices are approximate (early-2026, EU market). Equivalent alternatives are noted where relevant.

2.1 Controller (Raspberry Pi)

Component	Model	Approx. Price	Notes
Raspberry Pi 4B — 4 GB	RPi 4 Model B	€ 55 – 65	4 GB is sufficient; 8 GB if budget allows
Official RPi Power Supply	USB-C 5V 3A	€ 8 – 12	Do not cheap out — unstable power = SD corruption
MicroSD Card (32 GB+) - rating A2 at least	SanDisk Endurance	€ 8 – 12	Use a 'high endurance' card for long write cycles
Raspberry Pi Case (optional)	Argon NEO / any	€ 10 – 15	Passive cooling case recommended

💡 **Raspberry Pi 5 Alternative**

The Raspberry Pi 5 (€ 70-90) is fully supported and roughly 2× faster. It is overkill for this use case but future-proofs your setup if you plan to add cameras, more sensors, or additional integrations.

If you buy a Pi 5, also purchase the official active cooler (€ 5) as it runs hotter than the Pi 4.

On a personal note: A used Pi 5 can be purchased for 80$, since the devs of HomeAssistant are not interested in making it lightweight and are keen on adding features that add computational cost, Pi 5 is the best solution for a small price increase

2.2 Zigbee Coordinator (If using Zigbee devices else skip)

The Zigbee coordinator is a small USB dongle that acts as the bridge between your Zigbee sensors/devices and the Raspberry Pi. It is required even if you later buy WiFi sensors, because the recommended sensor below uses Zigbee.

Component	Model	Approx. Price	Notes
Zigbee USB Coordinator	SONOFF Zigbee 3.0 USB Dongle Plus (E)	€ 18 – 22	Best-in-class stability, uses EFR32 chip
USB Extension Cable	Any 50 cm USB-A male/female	€ 3 – 5	Keeps the dongle away from Pi RF noise

⚠️ **USB Interference**

Plug the Zigbee dongle into the USB 2.0 port (black, not blue) of the Raspberry Pi and use a short extension cable to move it a few centimetres away from the board. USB 3.0 (blue port) generates radio noise in the 2.4 GHz band that can disrupt Zigbee communication.

2.3 Temperature & Humidity Sensor

Component	Model	Approx. Price	Notes
Temp/Humidity Sensor	SONOFF SNZB-02D	€ 12 – 16	Zigbee, ±0.2 °C / ±2 % RH accuracy, E-ink display
Alternatives	AqaraTH-S2 / Aqara WSDCGQ11LM	€ 12 – 18	Same Zigbee protocol, slightly higher accuracy

💡 **Sensor Placement**

Mount the sensor at canopy height — roughly 20-25 cm from the top of your tallest plant.

Keep it away from direct airflow from fans, which causes erroneously low humidity readings.

Avoid placing it directly under grow lights — radiant heat can skew temperature readings.

2.4 Humidity Control Devices

For a small grow box (up to 1 m³), you do not need industrial humidifiers. Consumer-grade ultrasonic humidifiers and dehumidifiers plugged into smart switches are perfectly adequate.

Component	Model	Approx. Price	Notes
Smart Plug × 2	SONOFF S26R2ZB (Zigbee)	€ 10 – 14 each	Controls mains socket, Zigbee — no hub needed beyond your coordinator
Ultrasonic Humidifier	Any ~1-2 L tank, mist nozzle type	€ 15 – 30	Look for continuous-on mode (no auto-off after X hours)
Mini Dehumidifier	Eva-Dry EDV-1100 or similar	€ 35 – 55	Peltier-type, silent, adequate for small spaces

⚠️ **Humidifier Auto-Off Warning**

Many consumer humidifiers automatically shut off after 4-8 hours as a safety feature. This will silently break your automation. When purchasing, confirm the humidifier stays on as long as it has water and the plug is powered. If yours has an auto-off, you may be able to disable it by holding the power button for 5 seconds on some models.

2.5 Network

The Raspberry Pi connects to your home router via Ethernet (strongly preferred) or WiFi.

Component	Model	Approx. Price	Notes
Ethernet Cable	Cat 5e or Cat 6, any length	€ 2 – 8	Preferred over WiFi for reliability
WiFi (built-in)	RPi 4 onboard 802.11ac	No cost	Acceptable fallback if Ethernet not possible

2.6 Total Budget Estimate

Item Category	Estimated Cost (EUR)
Raspberry Pi 4B 4GB + accessories	€ 75 – 95
Zigbee coordinator + USB cable	€ 21 – 27
Temperature/humidity sensor	€ 12 – 16
Smart plugs × 2	€ 20 – 28
Humidifier + Dehumidifier	€ 50 – 85
Total	€ 178 – 251

3. Software Setup

3.1 Flashing Home Assistant OS to the SD Card

Home Assistant OS (HAOS) is a purpose-built Linux distribution that runs Home Assistant as its only job. It handles automatic updates, add-on management, and backups out of the box. We install it using the official Raspberry Pi Imager.

Steps

1. Download and install Raspberry Pi Imager from raspberrypi.com/software on your Windows, macOS, or Linux computer.
2. Insert your microSD card into your computer.
3. Open Raspberry Pi Imager. Click "Choose OS" → scroll to "Other specific-purpose OS" → "Home assistants and home automation" → "Home Assistant" → select the Raspberry Pi 4 image.
4. Click "Choose Storage" and select your SD card.
5. Click "Write". The imager downloads and flashes HAOS automatically. This takes 3-8 minutes.
6. Eject the SD card safely and insert it into your Raspberry Pi.
7. Connect the Zigbee dongle (using the USB extension cable) to a USB 2.0 port on the Pi.
8. Connect Ethernet and power. The Pi will boot and perform first-run setup for 5-10 minutes.

> 💡 **First Boot**
>
> After powering on, open a browser on any device on the same network and navigate to http://homeassistant.local:8123
>
> If that does not work, find the Pi's IP address from your router's admin panel and use http://<IP>:8123
>
> The first boot can take up to 10 minutes — be patient before assuming something went wrong.

3.2 Initial Home Assistant Configuration

Create your account

9. On the welcome screen, click 'Create my smart home'.
10. Enter a name, username, and password. Keep these safe — this is your admin account.
11. Set your location and timezone. This is important for the correct local time in automations.
12. Skip the device discovery step for now — we will add devices manually.

(Only if using Zigbee) Install the Zigbee Home Automation (ZHA) Integration

13. Go to Settings → Devices & Services → Add Integration.
14. Search for 'Zigbee Home Automation' and select it.
15. Home Assistant will auto-detect your SONOFF dongle. Confirm the device path (usually /dev/ttyUSB0 or /dev/ttyACM0).
16. Click Submit. ZHA will initialise your Zigbee coordinator — this takes about 30 seconds.

3.3 Pairing Your Devices

Pairing the SONOFF SNZB-02D Sensor

17. In Home Assistant: Settings → Devices & Services → Zigbee Home Automation → Add Device.
18. Put the sensor in pairing mode: hold the reset button for 5 seconds until the display shows 'Join'.
19. Home Assistant will discover the sensor automatically. Give it the entity ID: sensor.grow_room_temperature and sensor.grow_room_humidity.

Pairing the SONOFF S26R2ZB Smart Plugs

20. Open 'Add Device' in ZHA again.
21. Plug in the S26R2ZB and hold its button for 5 seconds until it flashes rapidly.
22. Home Assistant discovers it. Name the first plug switch.grow_room_humidifier and the second switch.grow_room_dehumidifier.

> ⚠️ **Entity ID Naming is Critical**
> The VPD Regulator configuration file references devices by their entity IDs. Use the exact names above (sensor.grow_room_temperature, sensor.grow_room_humidity, switch.grow_room_humidifier, switch.grow_room_dehumidifier) to avoid having to edit the YAML file. You can rename entities under Settings → Devices & Services → Entities.

3.4 Installing Docker and Deploying VPD Regulator

The VPD Regulator is delivered as a Home Assistant configuration file and a Docker Compose file that orchestrates the Home Assistant container (config file for HA must be in the config folder that is mapped to a volume that Docker uses for the HA container). On HAOS, you only need to copy the files and run one command.

Option A — Via the Home Assistant Terminal Add-on (Recommended) (need git login)

23. In Home Assistant: Settings → Add-ons → Add-on Store. Search for 'Terminal & SSH' and install it.
24. Open the Terminal add-on. You are now in a shell on your Raspberry Pi.
25. Clone the repository:

```
cd /config
git clone https://github.com/MarcoCrisafulli5/VPDRegulator.git
cp VPDRegulator/VPD_Regulator.yaml configuration.yaml
```

26. Restart Home Assistant: Settings → System → Restart.

Option B — Via Docker Compose (if running HA Container, not HAOS)

If you are running Home Assistant Container (Docker) on a standard Raspberry Pi OS instead of HAOS, use the provided docker-compose.yml:

```
git clone https://github.com/MarcoCrisafulli5/VPDRegulator.git
cd VPDRegulator
mkdir -p homeassistant/config
cp VPD_Regulator.yaml homeassistant/config/configuration.yaml
docker compose up -d
```

> 💡 **Which option should I choose?**
>
> If you flashed HAOS (following Section 3.1), use Option A. HAOS manages Docker internally and does not expose docker compose on the command line.
>
> If you installed Raspberry Pi OS and Docker manually (advanced users), use Option B.
>
> For most first-time users we strongly recommend HAOS + Option A.

Option C - Creating a config file and pasting the configuration

3.5 Verifying the Configuration

27. After restarting, go to Settings → Developer Tools → States.
28. Search for sensor.calculated_vpd. You should see a numeric value (e.g. 0.82).
29. Search for sensor.vpd_target_humidity. This is the humidity % the system is targeting right now.
30. If either sensor shows 'unavailable', check that your sensor entity IDs match exactly (see Section 3.3).
31. Manually test device control: search for switch.grow_room_humidifier in the States panel and toggle it. The plug's LED should respond.

4. Using the Dashboard

4.1 Accessing the VPD Dashboard

After the configuration loads, a pre-built dashboard is available at Settings → Dashboards → VPD Control, or directly via the sidebar if it is set as a home dashboard. It is also accessible on mobile by opening the Home Assistant companion app (iOS / Android) and navigating to the same dashboard.

4.2 Dashboard Overview

Dashboard Section	What it shows
Current Conditions	Live temperature, humidity, and calculated VPD
VPD Target Selector	Dropdown to choose your current growth stage
Calculated Target RH	The humidity % needed to hit your selected VPD at current temperature
Device Status	On/Off indicators for humidifier and dehumidifier
4-Hour History Charts	Trend lines for temperature, humidity, and VPD
Event Log	Last 20 automation trigger events with timestamps

4.3 Changing the Growth Stage

32. On the dashboard, locate the 'VPD Target' card.
33. Click the dropdown and select your current growth stage (e.g. 'Early Vegetative').
34. The system recalculates the target humidity immediately. The next automation cycle (within 30 minutes) will act on the new target.
35. To force an immediate update: Settings → Developer Tools → Services → search for script.vpd_control → Call Service.

5. Configuration Reference

5.1 Key Parameters

All configuration lives in VPD_Regulator.yaml (copied to configuration.yaml). The table below describes the most important parameters you may want to adjust.

Check also: readme chapter

Component	Model	Approx. Price	Notes
sensor.your_actual_temp_sensor	Entity ID	Your sensor	Replace with your real temperature entity ID
sensor.your_actual_humidity_sensor	Entity ID	Your sensor	Replace with your real humidity entity ID
rh_target - 2	Dead-band %	-2 %	Turn humidifier ON when humidity is 2 % below target
rh_target + 3	Dead-band %	+3 %	Turn dehumidifier ON when humidity is 3 % above target
minutes: /30	Interval	Every 30 min	How often the automation checks and acts

5.2 Adjusting the Dead Band

The dead band prevents your humidifier and dehumidifier from short-cycling (turning on and off every few minutes). A wider dead band means less device wear but allows humidity to drift further from the target. For a small tent with a responsive ultrasonic humidifier, ±2-3 % is a good starting point. If your devices are cycling too frequently, increase to ±5-7 %.

```
# In script.vpd_control → conditions block:
# Turn on humidifier:
value_template: "{{ current_rh < (rh_target - 2) }}"   # Change -2 to -5 for wider band
# Turn on dehumidifier:
value_template: "{{ current_rh > (rh_target + 3) }}"   # Change +3 to +7 for wider band
```

5.3 Adjusting the Check Interval

By default the automation runs every 30 minutes. For a very small space that reacts quickly to device changes, you may want to check every 10-15 minutes. For a more stable environment, every 60 minutes is fine.

```
# In automation → trigger block:
- platform: time_pattern
  minutes: "/15"   # Change to /10, /20, /30, /60 as needed
```

6. Design Decisions & Rationale

This section documents the key technical choices made during the design of the VPD Regulator and the reasoning behind them. If you are considering modifications, this context will help you make informed decisions.

6.1 Why Zigbee Instead of WiFi Sensors?

Factor	Zigbee vs WiFi Sensors
Network congestion	Zigbee uses 2.4 GHz sub-channels separate from WiFi; doesn't add to WiFi traffic
Power consumption	Zigbee sensors run for 12-24 months on 2 AAA batteries; WiFi sensors need USB power
Latency	Zigbee reports state changes in 1-2 seconds; WiFi varies from 1-10+ seconds
Reliability	Zigbee mesh self-heals if one node drops; WiFi sensors require good signal to AP
Cost	Comparable price point — both around € 12-18 per sensor
Local control	ZHA (local) vs cloud-dependent WiFi integrations; Zigbee wins on privacy

6.2 Why HAOS Over Bare Docker?

Home Assistant OS is a supervised environment that manages the HA process, add-ons (like the Terminal), automatic backups, and OS-level updates as a single unit. This is significantly easier to maintain than a bare Docker installation, where you are responsible for container orchestration, system updates, USB passthrough configuration, and backup scheduling separately. For end-users without Linux system administration experience, HAOS removes the majority of failure points.

The trade-off is that HAOS is less flexible — you cannot easily run arbitrary software alongside it on the same Pi. If you need the Pi for other tasks, use Raspberry Pi OS + Docker Compose (Option B in Section 3.4). For dedicated VPD control, HAOS is the right choice.

6.3 Why Dead-Band Control Instead of PID?

A PID (Proportional-Integral-Derivative) controller would in theory provide smoother humidity regulation, but for grow-room humidification it is impractical. Consumer humidifiers and dehumidifiers are binary devices — they are either on or off. There is no modulation of output. A PID loop would pulse them on and off rapidly ("bang-bang" behaviour at fine timescales), causing excessive wear. Dead-band control acknowledges the binary nature of the actuators and sets a hysteresis window that is matched to the slow humidity response time of the space (typically 5-20 minutes per 1 % RH change in a small tent).

6.4 Why Magnus-Tetens for VPD Calculation?

The Magnus-Tetens formula is an empirical approximation for saturation vapor pressure as a function of temperature. It is accurate to within 0.1 % over the temperature range 0-40 °C — more than sufficient for horticultural use. It is also simple enough to express in a single Home Assistant template string, avoiding the need for external Python scripts or integrations. More accurate equations (e.g. Buck or Arden Buck) exist but the marginal improvement does not justify the added complexity.

7. Troubleshooting

7.1 Quick Diagnostics

Symptom	First step
sensor.calculated_vpd shows 'unavailable'	Check sensor entity IDs in Settings → Developer Tools → States
Devices not switching on/off	Manually toggle switch in States panel to confirm device pairing
Automation not running	Settings → Automations → confirm 'VPD Control Automation' is enabled
Zigbee sensor offline	Check battery; move sensor closer to coordinator; re-pair if needed
Dashboard not visible	Settings → Dashboards — ensure 'VPD Control' dashboard exists
Home Assistant unreachable	Check Pi power supply; connect monitor to Pi to see console logs

7.2 Checking Logs

On HAOS, access logs via Settings → System → Logs. For more detail, enable debug logging:

```
# Add to configuration.yaml:
logger:
  default: info
  logs:
    homeassistant.components.script: debug
    homeassistant.helpers.template: debug
```

If running Docker (Option B), view logs with:

```
docker logs -f homeassistant-vpd
```

7.3 Humidity Not Reaching Target

- Confirm the humidifier does not have an auto-off feature (see Section 2.4 warning).
- Check that the smart plug entity switch.grow_room_humidifier turns the socket on (test manually in Developer Tools).
- If the tent is very dry (below 30 % RH), the humidifier may need to run for 20-40 minutes before humidity rises — the 30-minute check interval is appropriate.
- Consider reducing the dead-band lower threshold (e.g. from -2 % to -1 %) to trigger the humidifier earlier.

8. Advanced Features

8.1 Automated Stage Scheduling

If your grow follows a fixed light schedule (e.g. 18/6 vegetative, 12/12 flowering), you can automate the VPD stage changes. Add these automations to configuration.yaml:

```
automation:
  - alias: "Switch to Flowering VPD at Week 4"
    trigger:
      - platform: time
        at: "06:00:00"
    condition:
      - condition: template
        value_template: '{{ now().weekday() == 0 }}'  # Monday only
    action:
      - service: input_select.select_option
        target:
          entity_id: input_select.vpd_target
        data:
          option: "1.0"  # Early Flowering
```

8.2 Mobile Alerts

Install the Home Assistant companion app (Android / iOS) to receive push notifications when VPD goes out of range or a device goes offline. Example alert:

```
automation:
  - alias: "VPD Critical High Alert"
    trigger:
      - platform: template
        value_template: >
          {{ states('sensor.calculated_vpd') | float > 1.6 }}
    action:
      - service: notify.mobile_app_your_phone
        data:
          title: "🚨 VPD Too High"
          message: "VPD is {{ states('sensor.calculated_vpd') }} kPa — check your
dehumidifier!"
```

8.3 Data Logging to CSV

Log sensor data every 15 minutes for analysis or compliance. Home Assistant writes the file to the /config directory, which you can access via the built-in file editor or SFTP.

```
automation:
  - alias: "VPD Data Logger"
    trigger:
      - platform: time_pattern
        minutes: "/15"
    action:
      - service: notify.file
        data:
          filename: "/config/vpd_data.csv"
          message: >
            {{ now().isoformat() }},
            {{ states('sensor.temperature') }},
            {{ states('sensor.humidity') }},
            {{ states('sensor.calculated_vpd') }},
            {{ states('input_select.vpd_target') }}
```

9. Appendix

9.1 Repository Structure

```
VPDRegulator/
├── VPD_Regulator.yaml       # Main Home Assistant configuration
├── docker-compose.yml       # Docker Compose for HA Container deployment
├── README.md                # Quick-start guide
└── docs/
    └── VPD_Regulator_Guide.pdf   # This document
```

9.2 Useful Links

Home Assistant — Raspberry Pi Installation

ZHA — Zigbee Home Automation Integration

VPDRegulator GitHub Repository

9.3 Glossary

Term	Definition
VPD	Vapor Pressure Deficit — difference between actual and saturated air moisture, in kPa
RH	Relative Humidity — percentage of maximum possible moisture currently in the air
kPa	Kilopascal — unit of pressure used to measure VPD (1 kPa ≈ 0.01 atm)
Dead band	A tolerance window around the target value within which no control action is taken
Zigbee	A low-power, mesh wireless protocol for IoT devices operating at 2.4 GHz
ZHA	Zigbee Home Automation — the Home Assistant integration for Zigbee devices
HAOS	Home Assistant Operating System — purpose-built Linux distribution for HA

Term	Definition
Entity ID	The unique identifier for a device or sensor in Home Assistant (e.g. sensor.grow_room_temperature)
Magnus-Tetens	Empirical formula used to approximate saturation vapour pressure from temperature

Questions? Open an issue on the GitHub repository or contact Marco Crisafulli.

🌿 **Happy growing** 🌿